



Computer Communications and Networks

LAB TASK NO 3

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Semester: 6th Semester

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Submission Date: May 12, 2026

Lab Task: Static Routing Configuration in Cisco Packet Tracer

Objective

Configure a network using static routing so that all PCs in different networks can communicate with each other.

Scenario

You are a network engineer. You need to design and configure a network with:

2 Routers

2 Switches

Each switch connected to 2 PCs

Each LAN must be able to communicate with the other LAN using static routing.

Topology Requirement

Router R1 connected to Switch S1

Router R2 connected to Switch S2

R1 and R2 connected via serial link

Each switch has 2 PCs

IP Addressing Requirements

LAN 1:

Network: 192.168.1.0/24

Assign IPs to:

Router interface

2 PCs

LAN 2:

Network: 192.168.2.0/24

Assign IPs to:

Router interface

2 PCs

Router-to-Router Link:

Network: 10.0.0.0/30

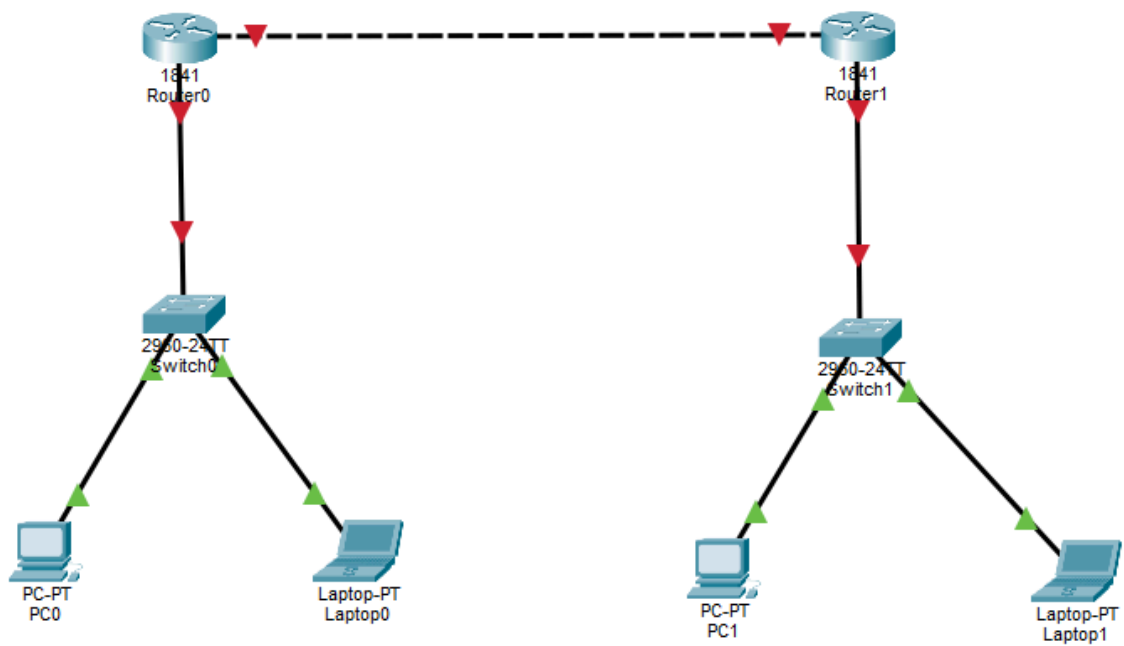
Assign IPs to both routers

Tasks

♦ Task 1: Create Topology

Add all required devices in Packet Tracer

Connect them using appropriate cables



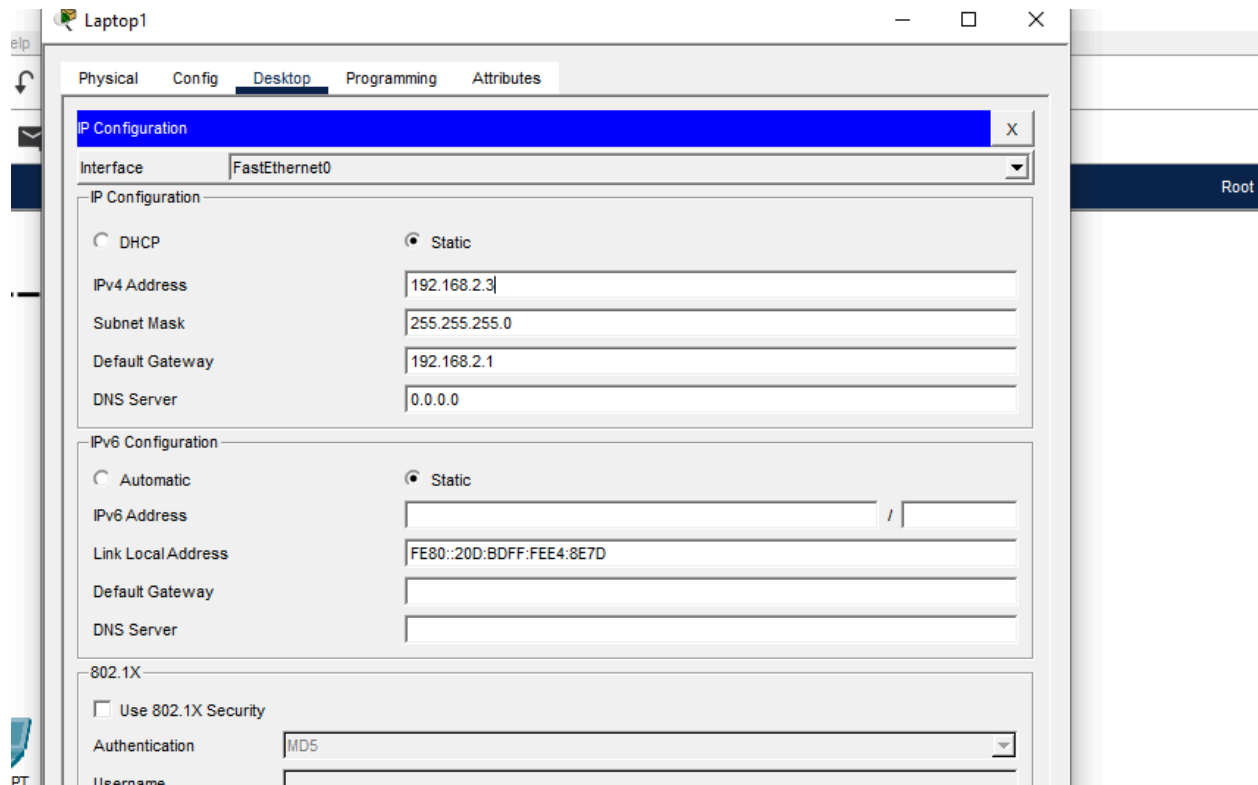
♦ Task 2: Configure IP Addresses

Assign IP addresses to:

Router interfaces

All PCs

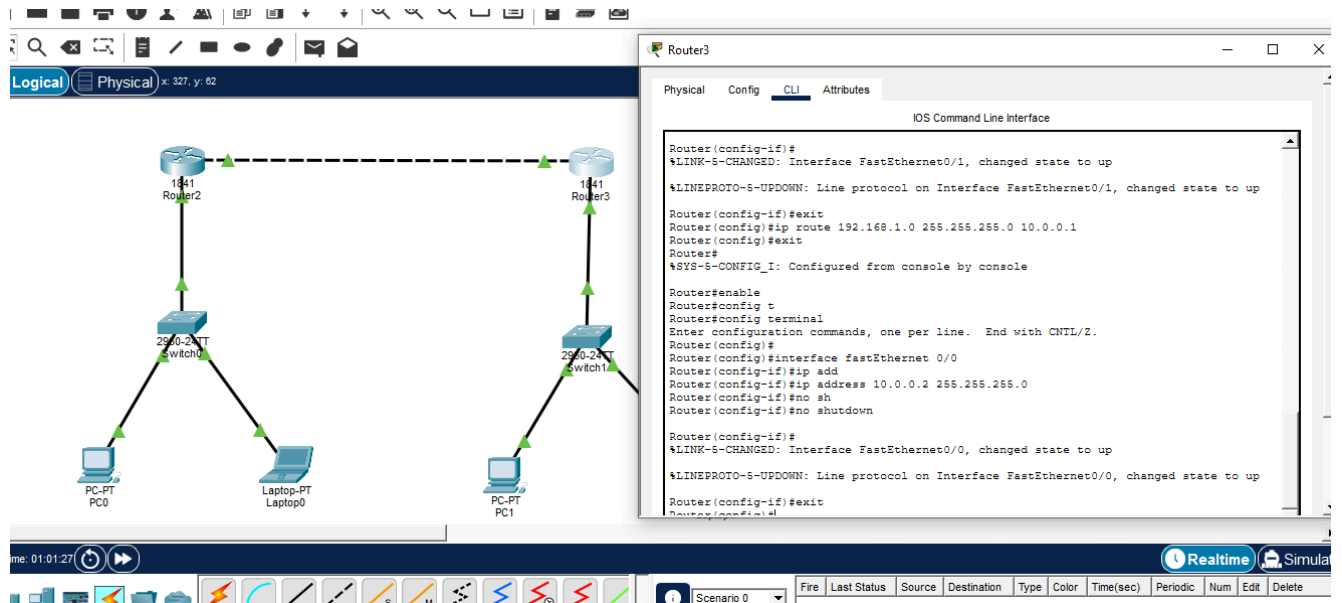
Configure default gateways on PCs



♦ Task 3: Configure Routers

Enable interfaces (no shutdown)

Configure serial link properly



◆ Task 4: Configure Static Routing

On R1, add route to LAN2

On R2, add route to LAN1

```

%SYS-5-CONFIG_I: Configured from console by console
Router(config-if)#exit
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

```

Top

◆ Task 5: Verification

Test connectivity using:

ping

Verify:

PC1 → PC3

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

PC2 → PC4

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

RealtimeSimulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Router2	Laptop0	ICMP		0.000	N	0	(e...	(delete)
	Successful	Laptop1	PC0	ICMP		0.000	N	1	(e...	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	2	(e...	(delete)
	Successful	Laptop0	Laptop1	ICMP		0.000	N	3	(e...	(delete)